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To cite this article: Yashaswi K Thakre and Prashant Y Pawade 2024 *IOP Conf. Ser.: Earth Environ. Sci.* **1326** 012094

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Traffic Congestion at Urban Road - Review

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ABSTRACT

One of the most obvious, ubiquitous, and urgent transportation issues that most cities throughout the world, including those in India, deal with on a daily basis is traffic congestion. All forms of mobility, particularly the roadways, are impacted, as well as all socioeconomic classes. Some of the main reasons of congestion are rapid population expansion, increased urbanization, bad public transportation systems, inadequate/unplanned transportation infrastructure, and an increase in the number of personnel vehicles. The results from research based on clogged roads are reviewed in this article. Numerous metrics to gauge traffic congestion have been considered. These matrices were divided into three categories: level of service, speed, and travel time. Additionally, methods for gathering data on congestion used in various researches have been reviewed. According to the study's conclusions, major factors in reducing congestion include better traffic management and control, better public transportation, increased investment for transportation infrastructure, the use of contemporary technology, and a general alignment of transportation and land-use policy.

Keywords: Traffic congestion, Traffic flow, Congestion measures, Data collection techniques

1. INTORDUCTION

The metropolitan is the one with the most traffic congestion, which has become a significant problem. Because of urbanization and the abundance of vehicles on the roads, the scenario of roadblocks becomes more serious with each passing day. Due to the convenience of 2-W and 4-W facilities, numerous private enterprises play a noteworthy share in the bottleneck in huge capitals. Urban and rural roadway congestion reduces productivity, loses valuable time and energy, augments psychological stress, and generates pollution that adds expenses to society and the country as a whole. Both medium-sized and small cities, as well as metropolitan areas, face the issue of traffic congestion. India is one of the most widely affected nations in the world by traffic halting, as per Tom (2022). Two of the top ten most populated cities in the world—Mumbai and Delhi —are located in India (2023 Ranking). The travel time of motorists of top cities in India like Delhi, Mumbai, Bengaluru, and Kolkata takes an average of one and a half hours more to cover the distance during peak hours of morning and evenings, according to a 2018 analysis by Boston Consulting Group. An Uber-commissioned BCG analysis found that India loses more than Rs 1.4 lakh crore annually with regard to travel time throughout times other than peak periods. Measurements of traffic overcrowding, **Stipancic et al.**, applying the traffic congestion index (CI) during peak hours demonstrates that there is an obvious connection between congestion and the frequency of accidents. 199 counties were included in a report on road fatalities and crashes as of 2018 by World Road Statistics. According to the report, India had the most fatal road accidents. It is believed that the initial phase in such tasks ought to be to identify the characteristics of the crowded portion of the road in order to find the most efficient alternatives. There are numerous explanations of traffic congestion presented by researchers or research organizations, however, they are entirely dependent on several traffic constraints, such as volume, capacity (or density), journey time (or lag), and speed. Though the claims of these thoughts hinged on statistics that have been collected, the study employs a variety of approaches to gather information. As a result, we can conclude that there is no single definition of congestion in the literature. (2004) Anthony. We have been

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using a study by Aftabuzzaman (2007) to determine the parameters for the density gauge. They propose the following criteria for a congestion measure: brevity, ease of use, stability, and consistency.
This paper's goal is to discuss various congestion control measures using both quantitative and qualitative metrics. Numerous procedures for measuring congestion at both the global and national levels have been thoroughly reviewed, and the results show numerous methods for gathering data. It will assist professionals, designers, and administrators in selecting an effective congestion gauge and carrying out a suitable information-gathering technique.

2. MEASURE OF TRAFFIC CONGESTION

Numerous measures have been created by various scholars to identify and quantify traffic congestion in n an effort to address the issue. These metrics can be useful for determining how congested the roads are and how well they are performing. These measurements are divided into three categories by us:
based on speed, based on travel time, and based on level of service.

2.1. Based on travel time: `

In studies of traffic congestion, travel time is the length of time it takes a vehicle to cross a certain stretch of road. Based on journey time, Urban Link's performance evaluation was done. The following is a list of some journey time metrics.

2.1.1. Delay:

In order to measure traffic congestion, delays are utilized. Delay can be characterized as extra travel time incurred while operating a vehicle in contrast to one's plans. Defining delay as the difference between average journey time (A_{jt}) and free-flow journey time (FF_{jt}), Lomax et al. Thus, equation (1) can be used to compute delay.

$$\textbf{Delay/Lag} = A_{jt} - FF_{jt} \quad (1)$$

2.1.2. Planning Time Index (PTI):

PTI is the proportion of the free-flow delay to the 95th % of journey time. (Karuppanagounder and Muneera, 2017). According to PTI, travel time also increases. PTI should therefore take the fewest possible actions to increase traffic flow.

$$\textbf{PTI} = \frac{\textbf{95\% JT}}{FF_{jt}} \quad (2)$$

2.1.3. Congestion Index (CI):

The proportion amid the delay and FF_{jr} is the CI metric. The difference between A_{jt} and FF_{jt} in this instance is delay. The A_{jt} for motorists would be kept to a minimum for optimal traffic management (Karuppanagounder and Muneera, 2017). The link's congestion index Roads that use travel time are

$$\textbf{CI} = \frac{\textbf{Delay}}{FF_{jt}} \quad (3)$$

2.1.4. Travel Time Index (TTI):

Rush-hour journey times (RH_{jt}) and free-flowing journey times (FF_{jt}) are compared using the TTI index (Lomax and Schrank, 2005). This approach has been applied in situations of ongoing and sporadic traffic congestion (Rao and Rao, 2012).

$$\textbf{TTI} = \frac{\text{RHjt}}{FF_{jt}} \quad (4)$$

2.1.5. Travel Time Index (TTI):

The ratio between A_{jt} and FF_{jt} is called TTI (Karuppanagounder and Muneera, 2017). The JTI should be kept to a minimum for improved traffic performance.

$$\textbf{JTI} = \frac{A_{jt}}{FF_{jt}} \tag{5}$$

2.1.6. Buffer Time Index (BTI):

The extra proportion of a commuter's duration must be on occasion in order for the buffer time index to accurately reflect 95% of the time (Nakat et al., 2014).

$$\textbf{BTI} = \frac{\textbf{95\%JT} - A_{jt}}{A_{jt}} \text{ X } 100\% \tag{6}$$

2.2. Based on Speed:

Speed is the most often used measure of efficiency for crowded roads and traffic. There are numerous approaches to calculate speed, starting with using the average trip time and the total length of the route under consideration. You may determine the average trip speed by

$$\textbf{Average Travel Speed} = \frac{\text{length of the route}}{\text{average trip time}} \text{ X } 100\% \tag{7}$$

Second, the stream's average speed is the weighted mean of the spot speeds of the vehicles that were detected by radar guns during the count time. The following formula is used to compute it at that location (Jain and Jain, 2017).

$$\textbf{Average Travel Speed} = \frac{\sum_{i=1}^{n} u_i . q_i}{\sum_{i=1}^{n} q_i} \tag{8}$$

Where, Vs = Mean speed of the traffic flow,
u_i = Mean spot speed of *i*th class of automobile,
qi = Stream of the *i*th group of automobiles and *n* is the number of classes.

2.2.1. Congestion Index (CI):

The congestion index depends on the speed utilized by Stipancic et al. (2016) for the estimation of congestion using Remote Sensing Information offers insights on network performance from both a microscopic and a macro level. The formula was used to calculate traffic congestion by Dias, et al.

$$\text{CI}=\begin{cases} \frac{V_F - V_A}{V_F}, & \text{If} \quad \text{CI} < 0 \\ 0 & \text{If} \quad \text{CI} \geq 0 \end{cases} \tag{9}$$

Where, VF = free-flow speed, and VA= actual speed.

2.2.2. Speed Reduction Index (SRI):

The speed reduction index compute the extent at which vehicle velocity is lowered as a effect of stream of traffic. By employing a constant as a gauge to differentiate among several categories of traffic overcrowding to different levels, this frequency enables linking the grade of traffic overcrowding for numerous sorts of conveyance amenities. As an effect, it can be utilized for several municipal street routes or precise street sections during rush or non-rush periods. (2015) Kukadapwar and Parbat. Following equation is used to evaluate it,

$$\textbf{SRI} = \frac{\text{Non peak flow speed} - \text{peak flow speed}}{\text{Non peak flow speed}} \tag{10}$$

2.2.3. Very-low-speed Index(VLSI):

The ratio between the length of time travelling at a considerably slower rate and the overall amount of time travelled is known as the very low-speed index. It has been applied to evaluate traffic congestion on urban streets (Kukadapwar and Parbat, 2015). Thus, the following equation can be used to obtain a very low-speed index.

$$\mathbf{VLSI} = \frac{\text{Time spent in delay}}{\text{Total travel time}} \quad (11)$$

2.2.4. Corridor Mobility Index (CMI):

The CMI is utilized to regulate a corridor's volume. A corridor mobility index is created by merging the velocity of passenger bulk and the par standards for road and major performance (Lomax, 1990). For high-occupancy vehicle lanes on roadways.

$$\boldsymbol{CMI_F} = \frac{\text{Travel speed(mph)} \times \text{peak-hour person volume per lane}}{100000} \quad (12)$$

For arterial high-occupancy-vehicle lanes

$$\boldsymbol{CMI_A} = \frac{\text{Travel speed(mph)} \times \text{peak-hour person volume per lane}}{20000} \quad (13)$$

2.2.5. Travel speed rate:

TSR is the proportion between the speed decrease in a congested area and the speed in a situation when there is no traffic congestion. 2002 (Hamad and Kikuchi).

$$\boldsymbol{TSR} = \frac{V_F - V_{AV}}{V_F} \quad (14)$$

Where VF = free-flow velocity,

VAv = average velocity

2.3. Based on Level of service (LOS):

LOS is commonly utilised as a form of qualitative evaluation to illustrate how the flow of traffic is working and how both motorists and passengers experience it. Traffic performance is identified as a threshold using LOS. LOS is typically divided into six levels, ranging from A to F. (1985 Manual). In which the LOSs for free-flow and congestion, or stop-and-go flow, respectively, are A and F, respectively. When operating speed is plotted with respect to the volume/capacity ratio for urban scenarios, LOS can be expressed at several levels.

2.3.1. Roadway Congestion Index (RCI):

Daily vehicle-mile travel per lane-mile (DVMT) for motorways and primary arterial street systems (PASS) is summed as a ratio of RCI, which evaluates the actual DVMT with the values estimated for crowded situation. This ranking enables assessments between entire neighborhoods in well-known cities by measuring the overall system performance and concentrating on the physical ability of the roadway as it relates to vehicles. The equivalence is used to determine it,

$$\boldsymbol{RCI} = \frac{(\text{Freeway DVMT/lane-mile}) \times \text{Freeway DVMT} + \left(\text{Arterial}\frac{\text{DVMT}}{\text{LANE}} - \text{MILE}\right) \times \text{FreewayDVMT}}{13000 \times \text{Freeway DVMT} + 5000 \times \text{Arterial DVMT}} \quad (15)$$

Consequently, once more on an area basis, 13,000 vehicles per lane per day for motorways and 5,000 vehicles per day for major arterial highways were used for congestion restrictions.

2.3.2. Congestion Severity Index (CSI):

Urban area roadway delay has been determined using CSI, which computes delays in travel time per million vehicle kilometers. The CSI employs a rate record to determine every hour of a typical day and the 1985 HCM computation and local motorway traffic count distribution to forecast repeating delays. On a different part of a motorway, Lindley (1987) conducted study using this measure and presented a criterion for overcrowding (v/c ratio of 0.77 or higher).

2.3.3. Lane Mile Duration Index (LMDI):

In order to quantify recurrent motorway congestion in urban networks, the LMDI was developed (Cottrell, 1991). Based on inputs from the Highway Performance Monitoring System (HPMS) to the roadway of the urban section, this index was utilised to construct an AADT/C. Equation can be used to compute LMDI.

$$\textbf{LMDI}=\sum_{j=1}^{N} \text{congested lanes mine}_j \times congested\ duration_j \quad (17)$$

Where, j = specific section of a motorway
N = The total amount of motorway sections in a city

3. LITERATURE REVIEW

Prabin et. al. studies that lacking in intersection infrastructure and increase in use of privately owned vehicle was main reason causing traffic congestion at Thirumazhisai junction. They have examined the study area with different methods of traffic volume counts. Data like vehicle classification, their moving pattern, direction of travel as well as pedestrian movement is collected by field survey. They suggested signal design improvement according to IRC: 106-1990 ―Guidelines on Vogue and Installation of Road Traffic signals. **Divya et. al.** evaluates the traffic problems and sustainable modification of road intersection at Ettumanoor, India. The vehicle occupancy and pedestrian traffic at the intersections were examined. the characteristics of the junction that caused delay problems are identified. The intersection's improper planning, a lack of traffic signals, and unlicensed parking, according to an analysis of the data gathered, are the main causes of traffic jams. They have provided controlling measures like alternative operation plan and signalization of intersection. **Alberto et. al**. have provided detailed information about type of intersections and their traffic congestion problem according to IRC guidelines. An attempt had been made by **Saurabh et. al** to reduce the traffic congestion at Polytechnic Roundabout, Bhopal, which is five legged rotary, having asymmetrical geometries with varied traffic flow. The objective of this research is to evaluate the present traffic condition, level of service of each road and provide a solution to overcome accident and delay. **Jingfei et. al.** have given application of Analytic Hierarchy Process (AHP), the traffic status of intersections will be evaluated and decided by Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) method. The results demonstrate that the framework is capable of to use the provided information to accurately assess the real-world traffic status of the junction and offer sources to make the right decisions for the traffic executives. The algorithm is validated by an actual example of urban highway junction traffic congestion. **Musqit et. al.** studied that Pantha chowk is very important link between South Kashmir and Jammu-Srinagar. They have serious problem of congestion due to valley roads which extended for hours and distance in kilometers. They focused on intersection improvement having six directional movement instead of road modification that meets at intersection for economic abd less time consuming solutions. Traffic volume study done on six directional traffic movement at intersection for seven days. They also done intersection delay study (stop time delay) as vehicle is waiting in queue to approach intersection. They had given solution option to provide roundabout

and design 3 phased signal system. **Muneera et. al.** studies that traffic congestion caused negative impact on transport sector which cause increase in transportation cost hence they evaluate traffic congestion cost at signalized intersection in state of Kerala(Tiruananthpuram), India. They have consider factors like traffic volume, traffic delay and passenger occupancy, this study shows private vehicles constitute average share of 75% of total traffic compared to other vehicles. If do so cost will be reduce by 38% and hence it is important to shift usage of public transport in indian cities to minimize traffic congestion. **Pratik et. al.** examine and debate the ability assessment of highways under diverse traffic conditions by many authors, and they evaluate and study the characteristics that determine the ability of highways as well as the factors that influence it. To improve the state of the road and the flow of traffic, this article can be utilised to analyse various techniques for assessing the capacity of roads. It outlines the procedure for estimating the two-lane road's capacity under mixed traffic situations, calculates the influencing elements that have an impact on that capacity, and suggests correction factors for all those conditions. Based on these modification parameters and under diverse traffic conditions, the ability of a road with two lanes is calculated. **Mankar et al.** studied the capacity of metropolitan highways under varied conditions. It might be challenging to predict traffic volume for capacity calculation on the street. The issue of measuring flow might be solved by making use of dynamic PCU values. The urban highways' carrying capacity is explored using the Green Shield model, and compare the outcomes using a model for microscopic simulation. The abrupt rise in the road's lane width is measured, and the findings indicate with the widening of the roads Road capacity is also increases. **Satish Chandra et al.** provide a method for determining the capacity of two-lane roads under a variety of traffic conditions. They also analyse the influencing factors that affect the highway ability and provide modification factors for each field condition. Based on these modification factors, the capacity of the highway has been established in a range of traffic limitations.

4. CONGESTION DATA COLLECTION TECHNIQUES

There are two categories of data gathering techniques: quantitative data and qualitative data. One or both of these strategies were employed by researchers for studies on traffic congestion that have appeared in the literature. Following details discusses several quantitative information collecting strategies.

4.1 Trajectory data:

Space-time profiles of every moving vehicle in a chosen area of a road segment are provided by information gathered from trajectory (Treiber et al., 2013). In this strategy, cameras or a plane may see the chosen section of traffic immediately over a tall structure. tracking the whereabouts of each car by collecting information over time from camera footage using different applications. This form of data collecting allows for the measurement of traffic density and lane modifications. Li et al. (2008) suggested a unique method for estimating traffic congestion on roadways in an open-world setting as seen from TV cameras mounted on poles or buildings. An technique for calculating traffic on time data gathered by visual photographic pictures was published by Palubinkas et al. in 2008. This method is based on the recognition of vehicles on a road section using change detection between two photos taken at close intervals. Hinz et al. (2007) provided a theoretical outline for planned dual channel radar satellite missions to monitor the highway system from sky and provided examples using actual data to highlight the advantages and disadvantages. Camera-based solutions are expensive for students to use for data collecting since they use complicated and error-prone methodologies and require automatic and reliable algorithms to monitor the cars. Additionally, the camera-based approach provides data a few hundred metres from a road stretch.

4.2. Cross Sectional Data:

Cross sectional data include microscopic data and macroscopic data.

4.2.1. Single-vehicle data or microscopic data:

Microscopic information is gathered on a particular stretch of road, and it can be obtained via sonar or laser barrier by laying pneumatic tubes across the road. Under the road surface, induction loops are frequently used to capture microscopic data. When the length of the vehicle is known, a single-loop sensor may be used to calculate the length at a given location on a road segment, passing the time of the automobile, traffic density, and speed. A set section of road, such as 1 m, divides two (or more) induction loops to form the double-loop detector. In order to divide vehicles into different categories based on axle spacing, this system requires two inductive loop detectors positioned at strategic locations along the route. (Cassady and Coifman, 2002). This technique of data collection is comprehensive and offers the distance travelled by cars between detection sites. An appealing, affordable replacement for inductive loops in traffic measurement on motorways and junctions is cordless magnets network of sensors (Cheung et. al., 2005). It is employed to determine occupancy, speed, and the quantity of cars. Based on the nonaxle, these sensors are utilised to classify the type of cars. Vehicle spacing that is not possible to calculate using conventional loop data.

4.2.2. Aggregated data or macroscopic data:

Average a collection of microscopic single-vehicle data across certain time periods will yield the macroscopic data, also known as aggregate data. The time period ranges from 20s to 5min, with 60s being the most typical interval. In 2020, Ranjan et al. suggested a different approach to gathering macroscopic data. South Korea's TOPIS (Seoul Transportation Operation and Information Service) is an open-source internet web service. It will employ a snapshot of the traffic congestion maps as an effective and affordable data collecting tool throughout the city. Similar to this, city-wide traffic congestion levels may be established in real-time by utilising several types of traffic maps from other nations, such as Google map, Bing map, and Baidu map.

4.3. Floating Car Data:

Data from floating cars only include information on one particular vehicle. These cars have specialised equipment. Floating data, as opposed to trajectory data, is recorded inside a vehicle. With this technique, GPS receivers may be used to get georeference coordinates. to a path on the road connect after being "map-matched" to it. It can calculate the distance travelled and the speed at two GPS stations in a road area. (Taylor, 1992). The probing vehicle was equipped with more sensors (such as infrared) that were used to measure the speed and distance travelled by the car in front (Stipancic et al.,
2017; Stipancic et al., 2019). Thus, extended floating-car data or enhanced floating-car data is used to describe it. Due to the abundance of equipped vehicles—such as taxis, vans, buses, and trucks—and their low speed on the highways, we are unable to fully depict the traffic using data from floating cars. Fortunately, this bias disappears when the information from floating automobile data becomes relevant: under free-flow speed and crowded settings, difference does not important.

5. DISCUSSION:

Congestion is primarily brought on by four factors: the atmosphere, equipment, people, and structure. The aforementioned scholars have discussed and recommended both quantitative and qualitative congestion indexes. The explanation of various congestion data gathering techniques will help in the choice of a method that is appropriate for the conditions in India. Depending on the budget, either a speed-based or LOS-based matrix can be used to quantify congestion.

When it comes to making the most efficient use of already-existing resources, such as private automobiles. The majority of the famous cities have lately contributed to this rise. The new point-to-point

moving concept known as carpooling is characterized by seven features: Several examples include flexible supply bases, dynamic pricing, dynamic routing, adaptive transporting, user network impact, demand pooling, feedback collecting, and management system.
In highway engineering, Congestion pricing is a transportation strategy that tries to relieve traffic jams in metropolitan zones by collecting forfeits for commuting in a specified area or hours of the day. In dynamic pricing, the cost is not predetermined in advance. Instead, it varies according to shifting variables, such as seasonal fluctuations in demand, the kinds of users being targeted, or variations in market dynamics. Via the application of computer programs, dynamic routing analyzes a variety of potential roads and chooses the most effective route for transport to follow throughout the whole system.

The advantages of car-pooling automatically decrease traffic jams by 17%-31% and maximize the investment in infrastructure. (1) Increasing the pace of public transportation; (2) offering alternatives to automobile ownership; (3) boosting earnings; (4) maximizing infrastructure placement and timing to automobile ownership, (5) Increasing earnings, and (6) Improving the timing and placement of infrastructure. Delhi will roll out a smartphone app based on a new-age technology linked to well-designed cameras located at signal lights to assist calm the hectic roadways (traffic cops are not required to physically control traffic). Signals will change from red to green in actual time depending on the amount of congestion, as opposed to altering at scheduled times. In Sydney, Australia, Singapore, Amsterdam, and other cities, the approach is currently being changed because of this study into successful models. At Closure Home in Coimbatore, Tamil Nadu, India, a similar system founded on real-time congestion is operational.

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